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IB Mathematics Application & Interpretation
Higher Level–Prediction Exams
November 2026 - Paper 2

Duration: 2 hrs (Calculator is Allowed) **Max. Marks: 110**

Question 1

[Maximum mark: 16]

A Ferris wheel has a diameter of 50 metres. The lowest point of the wheel is 2 metres above the ground. The wheel completes one full revolution every 4 minutes, and a seat begins its rotation at the lowest point.

(a) Find the distance travelled by a seat during one complete revolution. [2]

The height h metres of the seat above the ground, t minutes after the rotation begins, is modelled by

$$h(t) = a\cos(bt) + d.$$

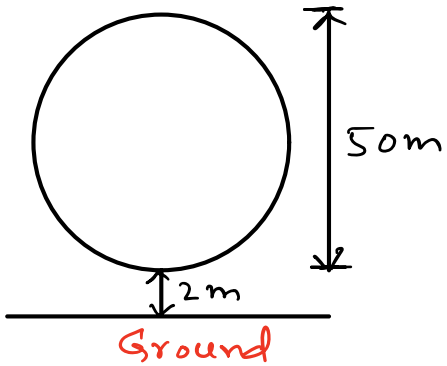
(b) (i) Write down the value of d.

(ii) Find the value of a.

(iii) Find the value of b. [6]

(c) Find the height of the seat after 1.5 minutes. [2]

Answer: Period = 4minutes



(a) $r = 25m$

Therefore, distance travelled in one rotation $= 2\pi r$

$= 50\pi$

$\approx 157\ m$

(b) $h(t) = a\cos(bt) + d$

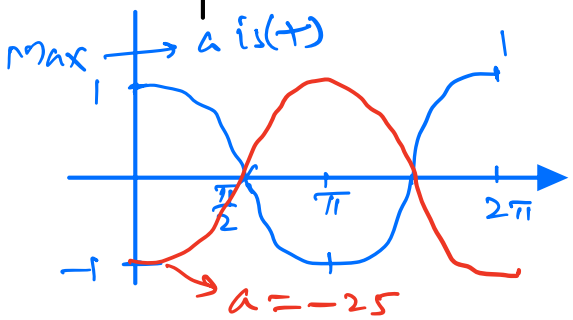
(i) $d = 27\ m$

(ii) $a = \left|\frac{Maximum - Minimum}{2}\right|$

$a = \left|\frac{52-2}{2}\right| = 25$

But wheel starts at lowest

$\therefore\ a = -25$



$$(iii)\ b = \frac{2\pi}{4} = \frac{\pi}{2}$$

(c) Find the height of the seat after 1.5 minutes. [2]

$h(t) = -25\cos\left(\frac{\pi}{2}t\right) + 27$

$h(1.5) = -25\cos\left(\frac{\pi}{2} \times \frac{3}{2}\right) + 27$

$= -25\left(-\frac{1}{\sqrt{2}}\right) + 27$

$h(1.5) = 44.67$

$\therefore$ height $= 44.7$ m

When a seat is higher than 40 metres, it is possible to view the local town centre.

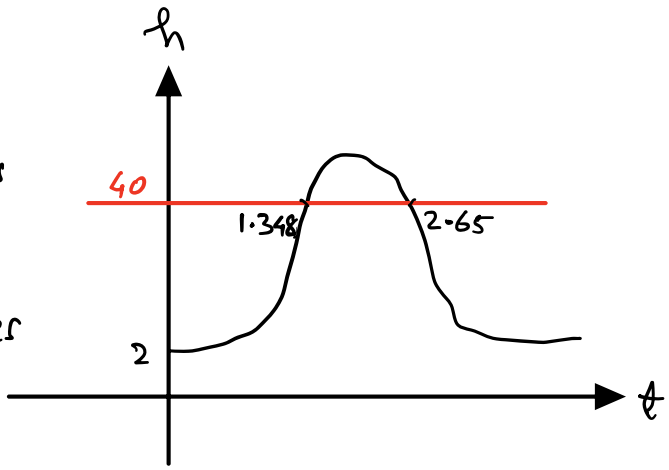
(d) Find the first time that the town centre is in view. [2]

A single ride on the Ferris wheel takes 20 minutes.

(e) Find the length of time, during a single ride, for which the town centre is in view. [4]

Answer:

(d) at $t =$ 1.35 minute



(e) length of single revolution is

$2.65 - 1.35 = 1.30$ min

One revolution takes 4 minutes

$\therefore$ In 20 minutes there will be 5 revolutions

$\therefore$ In 20 minutes the town centre is in view for

$\therefore\ 5 \times 1.30 =$ 65 minutes

Question 2

[Maximum mark: 15]

Alan has inherited a collection of 1000 stamps which he has started selling online. He records the number of stamps he sells each week.

In the first week he sells 10 stamps. In the second week he sells 12 stamps. In the third week he sells 14 stamps.

10, 12, 14

It can be assumed that the number of stamps Alan sells each week will follow an arithmetic sequence until all the stamps have been sold.

(a) Determine the number of stamps Alan has remaining after week 11. [3]

At the time Alan inherited the collection, the mean value of the stamps was \$2, but the mean value is increasing by 0.05% per week (compounding weekly).

(b) Calculate the total value of Alan's remaining stamps after week 11. Give your answer to the nearest dollar. [3]

(c) (i) Calculate the value of S_{28}.

(ii) State what this value means in context. [3]

Brian is a stamp collector. He starts buying stamps online in the same week as Alan starts selling stamps online. It can be assumed that the number of stamps Brian buys each week will follow an arithmetic sequence.

In the first week, Brian buys 2 stamps. After 3 weeks, Brian has bought 15 stamps in total.

(d) Calculate the common difference, d, for Brian's sequence. [2]

(e) Determine the first week in which Brian has bought more stamps in total than Alan has remaining in his collection. [4]

Answer:

(a) $10, 12, 14, \ldots$

$u_1 = 10, \quad u_2 = 12, \quad u_3 = 14$

$\therefore \quad u_2 - u_1 = 2, \quad u_3 - u_2 = 2$

$S_n = \frac{n}{2}[2u_1 + (n-1)d]$

$\therefore d = 2$

$S_{11} = \frac{11}{2}[2(10) + (11-1)(2)] = 220$ Sold

$\therefore$ Remaining stamps $= 1000 - 220 = 780$ stamps

(b) $FV = PV\left(1 + \frac{r}{100}\right)^n$

$= (780 \times 2)\left(1 + \frac{0.05}{100}\right)^{11} \approx \$1569$

(c) (i) $S_n = \frac{n}{2}[2u_1 + (n-1)d]$

$S_{28} = \frac{28}{2}[2(10) + (28-1)2]$

$\therefore S_{28} = 14(20 + 54) = 1030$ stamps

$c\,(ii)$ Alan has sold all his stamps as, 1030 > 1000.

$$(d)\ u_1 = 2, \quad S_3 = 15$$

$$\therefore \frac{3}{2}\left[2\,(2) + (2)\,d\right] = 15$$

$$4 + 2d = 15 \times \frac{2}{3} = 10$$

$$\therefore d = \frac{6}{2} = 3$$

(e) For Alan remaining stamps

$$1000 - S_n = 1000 - \left[\frac{n}{2}\,(2(10) + (n-1)2\right]$$

$$= 1000 - \left(9n + n^2\right)$$

For Brian total stamps collected in n weeks. $= \frac{n}{2}\left[2\,(2) + (n-1)\,3\right]$

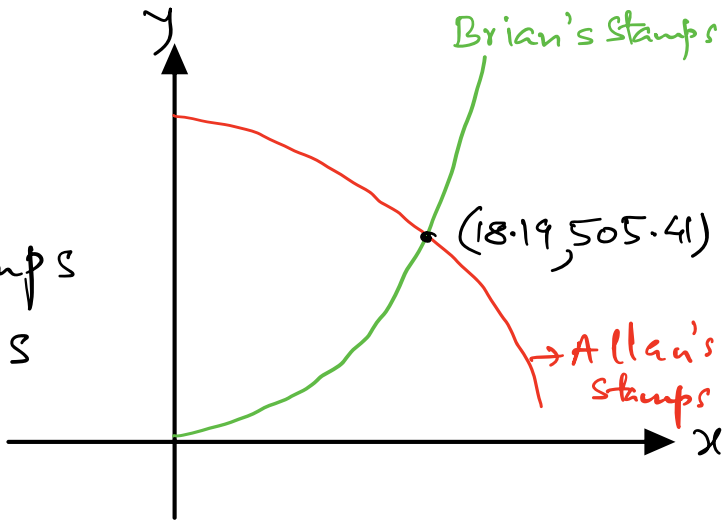
$$= \frac{3}{2}n^2 + \frac{n}{2}$$

$$\therefore \frac{3}{2}n^2 + \frac{n}{2} > 1000 - \left(9n + n^2\right)$$

Graphing both sides

$\therefore n = 19$

In week 19 Brian will have more stamps than remaining stamps of Alan.



Question 3

[Maximum mark: 16]

A factory produces bags of flour. The mass of a bag, M grams, is normally distributed with mean 500 g and standard deviation 8 g.

A bag weighing more than 510 g is classified as overweight.

(a) Find the probability that a randomly selected bag is classified as overweight. [2]

The lightest 10% of bags are classified as underweight

(b) Find the maximum mass of an underweight bag. [2]

Any bag not classified as overweight or underweight is cleared for distribution.

(c) Given that a bag has a mass of less than 515 g, find the probability that it is cleared for distribution. [4]

In one particular week, the weight classification does not occur and all bags are cleared for distribution.

A bakery orders a delivery of 15 bags during this week.

(d) Find the probability that at most 4 bags in this delivery should not have been cleared for distribution. [4]

The bakery in fact receives one delivery of 15 bags each day from Monday to Friday of this week. It can be assumed that each daily delivery is independent of the others.

The bakery will return an entire delivery if more than 4 of its bags should not have been cleared for distribution.

(e) Find the probability that the bakery will return at least one delivery during the week. [4]

Answer: $X \sim N\left(500, 8^2\right)$

$(a)\ P\left(x > 510\right) = 0.106 \rightarrow (0.10564977)$

$(b)\ P\left(x < m\right) = 0.10$

Using inverse normal on GDC

$$m = 490g$$

(c) 10% are underweight
& 10.6% are overweight

$$P\left(489.70 < x < 510 | x < 515\right) = \frac{P\left(489.70 < x < 510\right)}{P\left(x < 515\right)} = \frac{0.79539}{0.96960363} \approx 0.819$$

(d) we are given that 10% bags are under weight.

And we found probability of a bag being overweight is 0.106

Therefore, total probability of clarification not occurring is

$$0.10 + 0.10564977 = 0.20564977 = p$$

Let y be the number of bags that should not be cleared

$$\therefore Y \sim B\left(15, 0.20564977\right)$$

$$\therefore P\left(Y \leq 4\right) = 0.821 \qquad (0.82088833)$$

The bakery in fact receives one delivery of 15 bags each day from Monday to Friday of this week. It can be assumed that each daily delivery is independent of the others.

The bakery will return an entire delivery if more than 4 of its bags should not have been cleared for distribution.

(e) Find the probability that the bakery will return at least one delivery during the week. [4]

Answer:

Let R be the number of deliveries returned during the week

Probability of more than 4 bags being returned is

$$p = 1 - 0.82088833 = 0.17911167$$

$$\therefore R \sim B\left(5, 0.17911167\right)$$

$$P\left(R \geq 1\right) = 0.627$$

Question 4

[Maximum mark: 13]

A study models the monthly travel choices of 1000 commuters, each of whom travels either by **bus** or by **train**. Each month, 15% of bus users switch to the train, and 40% of train users switch to the bus. The monthly changes are described by the transition matrix

$$T = \begin{pmatrix} a & 0.40 \\ 0.15 & b \end{pmatrix},$$

where the columns represent the current month and the rows the following month. Initially 600 commuters travel by bus and 400 travel by train.

(a) (i) State the value of a and the value of b.

(ii) Hence find the number of commuters travelling by bus after 2 months. [3]

(b) Find the eigenvalues of T. [3]

(c) Find the eigenvectors corresponding to these eigenvalues. [3]

(d) Hence find the number of commuters expected to travel by bus in the long term. [2]

(e) Explain how the smaller eigenvalue confirms that the system approaches this long-term state. [2]

Answer:

(a)

		From	
		BUS	Train
To	Bus	0.85	0.40
	Train	0.15	0.60

(i) $a = 0.85$, $b = 0.60$

$\therefore T = \begin{pmatrix} 0.85 & 0.40 \\ 0.15 & 0.60 \end{pmatrix}$

$S_0 = \begin{pmatrix} 600 \\ 400 \end{pmatrix}$

(ii) After 2 months $\qquad S_n = T^n S_0$

$$S_2 = T^2 S_0$$

$$= \begin{pmatrix} 0.85 & 0.40 \\ 0.15 & 0.60 \end{pmatrix}^2 \begin{pmatrix} 600 \\ 400 \end{pmatrix} = \begin{pmatrix} 701.5 \\ 298.5 \end{pmatrix}$$

$\therefore$ 702 commuters travel by bus

(b) Eigen values are solutions to

$$\det(T-\lambda I)=0$$

If $A=\begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow \det A = |A| = ad-bc$

$T=\begin{pmatrix} 0.85 & 0.40 \\ 0.15 & 0.60 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$

$\therefore \det(T-\lambda I)=0$

$$\left|\begin{pmatrix} 0.85-\lambda & 0.40 \\ 0.15 & 0.60-\lambda \end{pmatrix}\right| = 0$$

$$(0.85-\lambda)(0.60-\lambda)-(0.15)(0.40)=0$$

$$\lambda^2-1.45\lambda+0.51-0.06=0$$

$$\lambda^2-1.45\lambda+0.45=0$$

$$\lambda_1=0.45 \ , \ \lambda_2=1$$

(c) For $\lambda_1=0.45$ Solving for $\vec{V}=\begin{pmatrix} x \\ y \end{pmatrix}$

$$(T-\lambda I)\vec{V}=\vec{O}$$

$$\begin{pmatrix} 0.40 & 0.40 \\ 0.15 & 0.15 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$0.40x+0.40y=0$$

$$x=-y$$

choosing $x=1$, eigenvector is

$$\vec{v}_1=\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

For $\lambda_2=1$

$$(T-\lambda I)\vec{V}=\vec{O}$$

$$\therefore \begin{pmatrix} -0.15 & 0.40 \\ 0.15 & -0.40 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$-0.15x+0.40y=0$$

$$x=\frac{0.40}{0.15}y \Rightarrow x=\frac{8}{3}y \Rightarrow \vec{V}_2=\begin{pmatrix} 8 \\ 3 \end{pmatrix}$$ is eigenvector

(d) Hence find the number of commuters expected to travel by bus in the long term. [2]

Using the fact that long term state is proportional to the eigenvector associated with $\lambda_2 = 1$, and scaling so that total is 1000, we have

For $\lambda_2 = 1$, $\vec{v}_2 = \begin{pmatrix} 8 \\ 3 \end{pmatrix}$

$$\text{Bus} = \frac{8}{8+3} \times 1000 = 727.2\ldots \approx 727 \text{ commuters}$$

(e) Explain how the smaller eigenvalue confirms that the system approaches this long-term state. [2]

Consider the power formula for a matrix (1.15)

$$M^n = PD^nP^{-1}$$

where D is diagonal matrix of eigenvalues.

In a transition matrix one of the eigenvalues will always be 1 and one will be with $|\lambda_n| < 1$.

In this case, we have

$$D^n = \begin{pmatrix} 0.45 & 0 \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 0.45^n & 0 \\ 0 & 1^n \end{pmatrix} = \begin{pmatrix} 0.45^n & 0 \\ 0 & 1 \end{pmatrix}$$

Now as $n \to \infty$, $0.45^n \to 0$

$$\therefore D^n = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Hence the contribution of this eigenvector dies away and the state converges to the steady state given by $\lambda = 1$ eigenvector.

Question 5

[Maximum mark: 14]

The population of a colony of penguins can be modelled by the differential equation

$$\frac{\mathrm{d}P}{\mathrm{d}t} = 66528e^{-0.0672t}\left(\frac{P}{10000}\right)^2, \quad t \geq 0,\ P > 0,$$

where P represents the number of penguins in the colony at time t years.

Initially, the colony was estimated to have 100 penguins.

(a) Determine the initial growth rate of the colony. [2]

(b) Show that the number of penguins present after t years can be expressed as

$$P = \frac{10000}{1 + 99e^{-0.0672t}}.$$

[7]

(c) Determine the maximum capacity of the penguin colony according to the model, justifying your answer. [2]

When the colony reaches 75% of its maximum capacity, some penguins leave the colony to find better access to food.

(d) Determine how long it will take for this to occur. Give your answer to the nearest year. [3]

Answer:

(a) When $t=0$, $\frac{dP}{dt} = 66528e^{0}\left(\frac{P}{10,000}\right)^2$

$$\frac{dP}{dt} = 6.6528 \quad \text{penguins per year}$$

(b) $$\int P^{-2}dp = \frac{66528}{(10000)^2}\int e^{-0.0672t}dt$$

$$\frac{1}{P} = \frac{66528}{(10000)^2}\frac{e^{-0.0672t}}{(0.0672)} + C$$

Using initial condition $P=100$ when $t=0$

$$\frac{1}{100} = \frac{66528}{(10000)^2} \times \frac{1}{(0.0672)} + C$$

$$C = 0.0001$$

$$\therefore \frac{1}{P} = 0.0099e^{-0.6672t} + 0.0001$$

$$\therefore \frac{1}{P} = 0.0099e^{-0.0672t} + 0.0001$$

Multiplying both sides by 10000 & rearrage

$$\frac{10000}{P} = 99e^{-0.0672t} + 1$$

$$P = \frac{10000}{(1 + 99e^{-0.0672t})}$$ ✓

(c) Determine the maximum capacity of the penguin colony according to the model, justifying your answer. [2]

As $t \to \infty$, $e^{-0.0672t} \to 0$

$\therefore$ $P = 10000$ penguins

When the colony reaches 75% of its maximum capacity, some penguins leave the colony to find better access to food.

(d) Determine how long it will take for this to occur. Give your answer to the nearest year. [3]

75% of maximum capacity is 7500

$\therefore P = 7500$

$\therefore$ from $P = \frac{10000}{(1 + 99e^{-0.0672t})}$

$$7500 = \frac{10000}{1 + 99e^{-0.0672t}}$$

using GDC

$t = 84.728\ldots$

$t = 85$ years.

Question 6

[Maximum mark: 17]

The concentration C, measured in mg/L, of a drug in a patient's bloodstream is recorded t hours after the drug is administered. The results are shown in the table.

t (hours)	1	2	3	4	5
C (mg/L)	7.8	6.1	4.7	3.7	2.9

The concentration is modelled by $C = ab^t$, where a and b are positive constants.

(a) Find an expression for $\ln C$ in terms of $\ln a$, $\ln b$ and t. [2]

(b) (i) Find the equation of the regression line of $\ln C$ on t.

(ii) Write down the value of Pearson's product-moment correlation coefficient, r, for $\ln C$ and t. [4]

(c) Find the value of a and the value of b. [4]

(d) Interpret the value of b in the context of the model. [1]

Patients are recommended to take a second dose of the drug after 8 hours, but not to take more than 2 doses in total.

(e) Use the model to estimate the concentration of the drug in the bloodstream immediately before the second dose is taken. [2]

The drug is effective while the concentration is at least 1 mg/L.

(f) Calculate how long the drug remains effective after the second dose is taken, assuming that the drug continues to deplete at a constant rate. Give your answer to the nearest minute. [4]

Answer: (a)

$$C = ab^t$$

$$\therefore \ln C = \ln\left(ab^t\right) = \ln a + \ln b^t$$

$$\ln C = (\ln b)\, t + \ln a$$

(b) (i) We have to calculate ln C from each data value, using above equation.

t (hours)	1	2	3	4	5
ln C (mg/L)	ln 7.8	ln 6.1	ln 4.7	ln 3.7	ln 2.9

Now we can find regression line of lnC on t using GDC, for this, we have

Menu → Stat → Stat Calc → 3 Line Reg(mx+b)

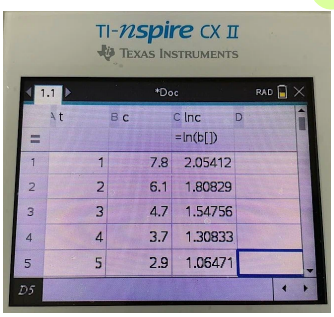
$$m = -0.2478, c = 2.30024$$

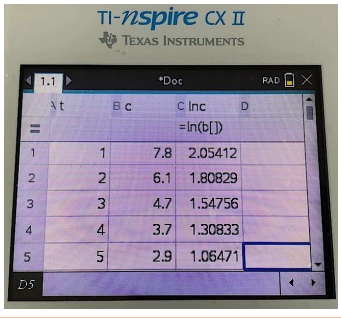
Hence equation of regression line of lnC on t is

$$\ln C = -0.248t + 2.30$$

(ii) We have correlation coefficient, r, from part (i)

$$r = -0.9999 \rightarrow r = -1$$





(c) Comparing $\ln C = \ln(b)t + \ln(a)$

with expression found in part (b)(i)

y-intercept $= \ln a$

$m = \ln(b)$

$\therefore \quad \ln a = 2.30024 \Rightarrow a = e^{2.30024}$

$a = 9.98$

and $\ln b = -0.2478 \Rightarrow b = e^{-0.2478}$

$\therefore \quad b = 0.780$

(d) As $b = 0.780 < 1$

and each hour concentration is multiplied by 0.780

So concentration decreases by approximately 22% per hour.

Patients are recommended to take a second dose of the drug after 8 hours, but not to take more than 2 doses in total.

(e) Use the model to estimate the concentration of the drug in the bloodstream immediately before the second dose is taken. [2]

Answer:

Patients are recommended to take second dose of drug after 8 hours i.e. $t = 8$

Using given formula

$C = a\,b^t$

$= 9.98\,(0.780)^8$

$C \approx 1.37 mg/L$

The drug is effective while the concentration is at least 1 mg/L.

(f) Calculate how long the drug remains effective after the second dose is taken, assuming that the drug continues to deplete at a constant rate. Give your answer to the nearest minute. [4]

Answer:

Second dose will add new concentration of drug to the bloodstream, which will be concentration at t = 0 , to the amount of already in the bloodstream found in part (e).

We have

$$\text{Concentration after 2nd dose} = 1.373\ldots + 9.976\ldots(0.7804)^0$$

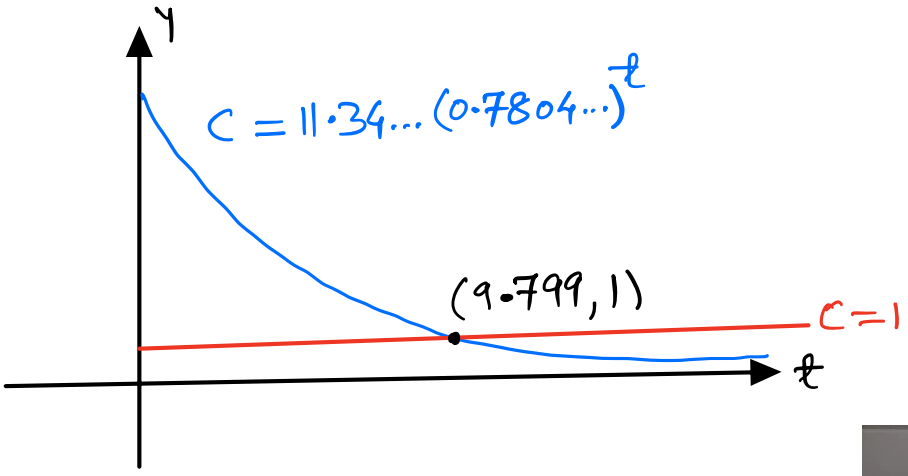
$$= 11.34mg/L$$

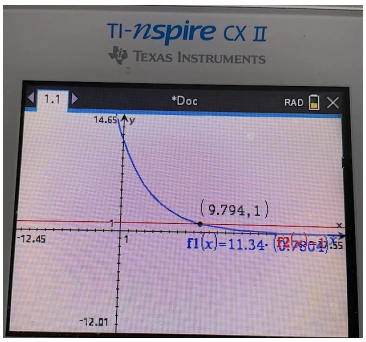
Assuming the same constant rate of 0.7804 mg/L per hour, our model for concentration of the drug in the bloodstream from the time the second dose is taken is

$$c = 11.34\ldots(0.7804\ldots)^t$$

We need to calculate when it falls below 1 mg/L.

Using graph to sketch C = 11.34..(0.7804..)^t and $C = 1$





The drug will not be effective after 9.799... hours.

t = 9 hours and 48 minutes

Question 7

[Maximum mark: 19]

A remote-controlled car moves across a flat, horizontal car park. Its position vector, in metres, relative to a fixed origin O, at time t seconds ($t \geq 0$) is modelled by

$$\mathbf{r}(t) = \begin{pmatrix} t^3 - 6t^2 + pt \\ q(t+1)\ln(t+1) \end{pmatrix}$$

where p and q are positive constants.

(a) Find the velocity vector $\mathbf{v}(t)$ at time t. [3]

(b) Find the acceleration vector $\mathbf{a}(t)$ at time t. [2]

At time $t = 0$, the car's speed is 10 m s^{-1}, and it is moving in the direction $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$.

(c) Find the value of p and the value of q. [5]

(d) (i) Find the value of t, for $t > 0$, at which the acceleration vector is parallel to the y-axis.

(ii) Find the angle between the velocity vector and the acceleration vector at this value of t.

It is given that

- If the angle between a velocity vector and an acceleration vector is acute, the object is speeding up.
- If a velocity vector and an acceleration vector are perpendicular, an object is moving at constant speed.
- If the angle between a velocity vector and an acceleration vector is obtuse, the object is slowing down.

(iii) Hence state, with a reason, whether the car's speed is increasing or decreasing at this instant. [9]

Answer:

(a) Differentiating displacement vector gives velocity vector.

$$\mathbf{r}(t) = \begin{pmatrix} t^3 - 6t^2 + pt \\ q(t+1)\ln(t+1) \end{pmatrix}$$

$$x(t) = t^3 - 6t^2 + pt \Rightarrow \therefore x'(t) = 3t^2 - 12t + p$$

$$y(t) = q(t+1)\ln(t+1)$$

$$y'(t) = q\left[(1)\ln(t+1) + (t+1)\frac{1}{(t+1)}\right] \quad \text{(Using product rule)}$$

$$y'(t) = q(\ln(t+1) + 1)$$

$$\therefore v(t) = \begin{pmatrix} 3t^2 - 12t + p \\ q(\ln(t+1) + 1) \end{pmatrix}$$

(b) $a(t) = v'(t) = \begin{pmatrix} 6t - 12 \\ \frac{q}{(t+1)} \end{pmatrix}$

(c) At $t = 0$, $v(0) = \begin{pmatrix} 3(0)^2 - 12(0) + p \\ q(\ln(0+1) + 1) \end{pmatrix} = \begin{pmatrix} p \\ q \end{pmatrix}$

Now magnitude of velocity is speed.

$$|v| = \sqrt{v_1^2 + v_2^2} = \sqrt{p^2 + q^2}$$

$$\sqrt{p^2 + q^2} = 10$$
$$\therefore p^2 + q^2 = 100$$

As car is moving in the direction $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$

So $\begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} 3s \\ 4s \end{pmatrix}$ for some $s \in R^+$

$$\therefore (3s)^2 + (4s)^2 = 100$$
$$9s^2 + 16s^2 = 100$$
$$25s^2 = 100$$
$$\therefore S = \frac{10}{5} = 2$$

$$\therefore p = 6, q = 8$$

(d)(i) A vector is parallel to the y-axis if its x component is zero.

From part(b) using acceleration vector, we have

$$\begin{pmatrix} 6t - 12 \\ \frac{q}{t+1} \end{pmatrix} = \begin{pmatrix} 0 \\ k \end{pmatrix}$$ for some $k \in R$

$$\therefore 6t - 12 = 0 \Rightarrow t = 2$$

(ii) Substituting $t=2$, $p=6$, $q=8$ in $v(t)$, we have

$$v(2) = \begin{pmatrix} 3(2)^2 - 12(2) + 6 \\ 8(\ln(3) + 1) \end{pmatrix} = \begin{pmatrix} -6 \\ 16.788\ldots \end{pmatrix}$$

Substituting $t=2$ & $q=8$ in $a(t)$, we have

$$a(2) = \begin{pmatrix} 0 \\ 2.66\ldots \end{pmatrix}$$

Using angle between two vectors formula (3.10), we have

$$\cos\theta = \frac{v_1 w_1 + v_2 w_2}{|v|\,|w|}$$

$$\cos\theta = \frac{(-6)(0) + (16.788)(2.666)}{\sqrt{(-6)^2 + (16.788)^2}\sqrt{0^2 + (2.666)^2}}$$

$$= 0.94167\ldots$$

$$\therefore \theta = \cos^{-1}(0.94167\ldots) \approx 19.7^\circ$$

(iii) As $0 < \theta < 90^\circ$ the angle between velocity vector and acceleration is acute

Therefore car's speed is increasing at t = 2.

End of Paper

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